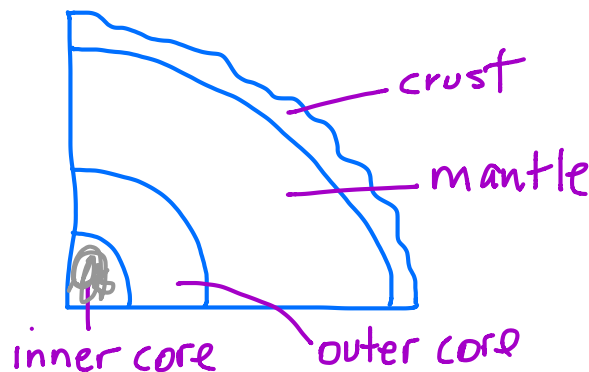


**Year 8 Science
Earth and Space Sciences
Revision - The Dynamic Earth**

1. (a) Which layers of the Earth have the following characteristics?

- Completely molten *outer core*
- Partially molten *mantle*
- Includes solid rock, soil and landforms *crust*
- Solid and mostly made up of iron *core*
- Lies above the surface *atmosphere*

(b) Draw a simple labelled diagram showing the layers of the Earth.



2. Describe two pieces of evidence that supported Wegener's theory of continental drift.

- *The coasts of continents fit together.*
- *Similar fossils are found on continents far apart.*

3. According to the theory of plate tectonics, the Earth's crust is divided into a number of slowly moving plates.

(a) What makes the plates move?

Convection currents in the mantle push the plates around.

(b) What can happen when two plates slide past each other?

Earthquakes occur.

- (c) How does the plate tectonics theory explain the increasing height of the Himalayas?

Two plates are pushing against each other, forcing land upwards to form mountains.

4. Where on Earth is the Ring of Fire and why does it exist?

- Around the edge of the Pacific Ocean.
- These are the edges of the continental plates where volcanoes occur.

5. How is an ocean ridge different from a subduction zone?

- Ocean ridge - plates moving apart, magma rises to form new land.
- Subduction zone - one plate moves under another and land is disappearing.

6. When oceanic crust pushes against continental crust, why does the oceanic crust slide underneath the continental crust?

Continental crust is less dense and "floats" above the oceanic crust as it sinks.

7. Describe the movements in the Earth's crust that cause the folding of rock that has shaped most of the Earth's mountains.

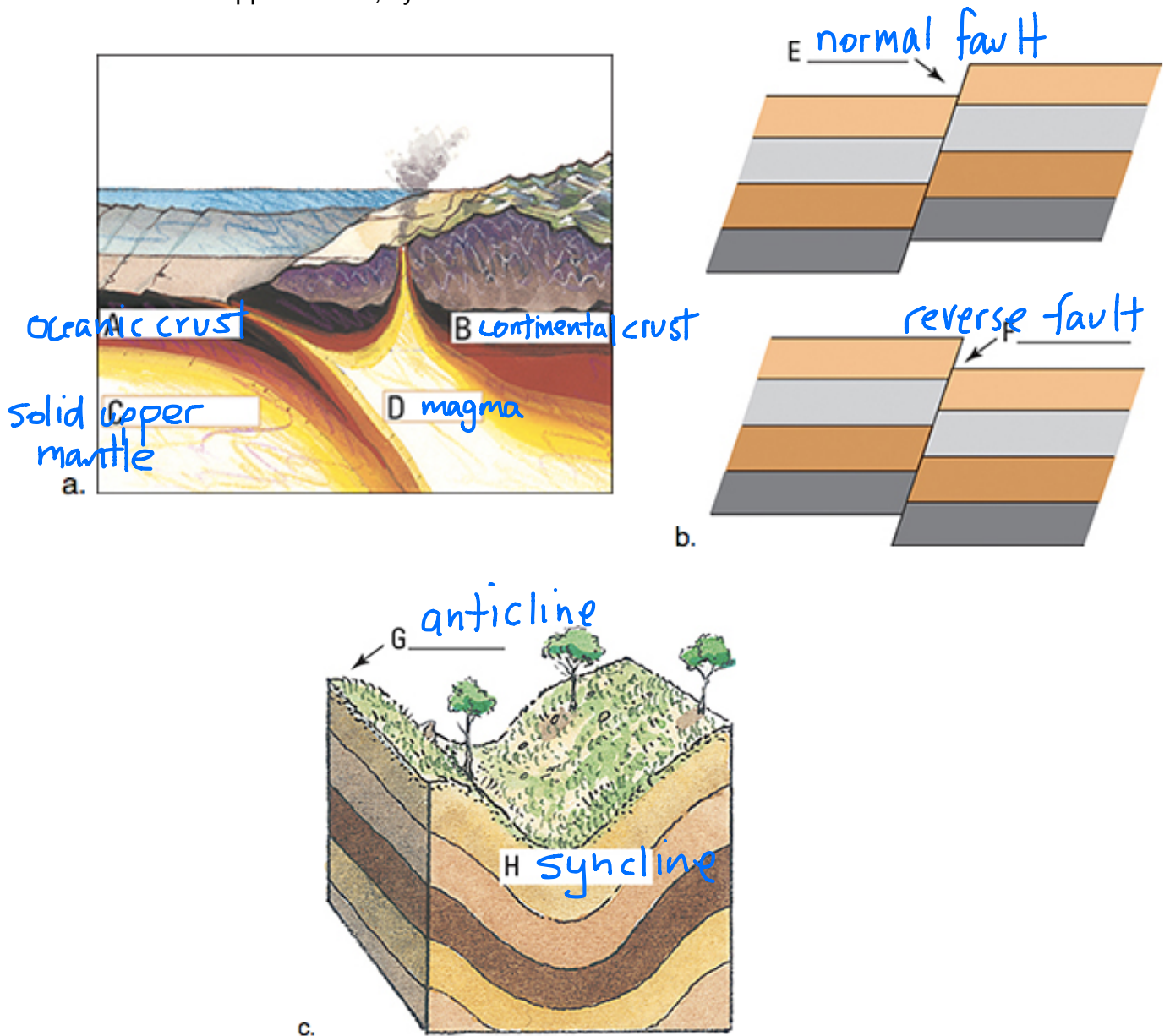
Plates push together and the layers of rock bend rather than breaking, causing a fold.

8. Explain how faults are created.

A fault is a large crack in the rock.
Land on either side starts to move.

9. Label the following diagrams using the following words:

anticline, continental crust, magma, normal fault, oceanic crust, reverse fault, solid upper mantle, syncline.



- (a) A subduction zone
 (b) Two types of faulting
 (c) Folding upwards and downwards

10. The San Andreas Fault makes much of coastal California, including the cities of Los Angeles and San Francisco, susceptible to earthquakes.

(a) Explain why the San Andreas Fault is called a slip fault.

Blocks of crust slip horizontally past each other.

(b) What causes major earthquakes along this fault?

Build up of pressure along the fault is suddenly released.

11. Distinguish between the epicentre of an earthquake and its focus.

Focus – point in the crust where the earthquake begins

Epicentre – point on the surface above the focus.

12. What is a seismograph used to measure?

The intensity (energy) of the earthquake.

13. Explain why Australia is less likely to experience volcanic activity and major earthquakes than New Zealand.

Australia lies in the middle of a plate so there are few earthquakes. New Zealand is on the edge of a plate so it has lots of volcanic activity.

14. Before a volcano erupts, its vents are blocked with thick, pasty lava.

(a) What change takes place to cause the volcano to erupt?

- Gases build up – pressure rises.
- Volcano bulges + then explodes.

(b) How is the lava emerging from a volcano different from magma?

Lava is a mixture of magma and various gases.

15. Distinguish between a dormant volcano and an extinct volcano.

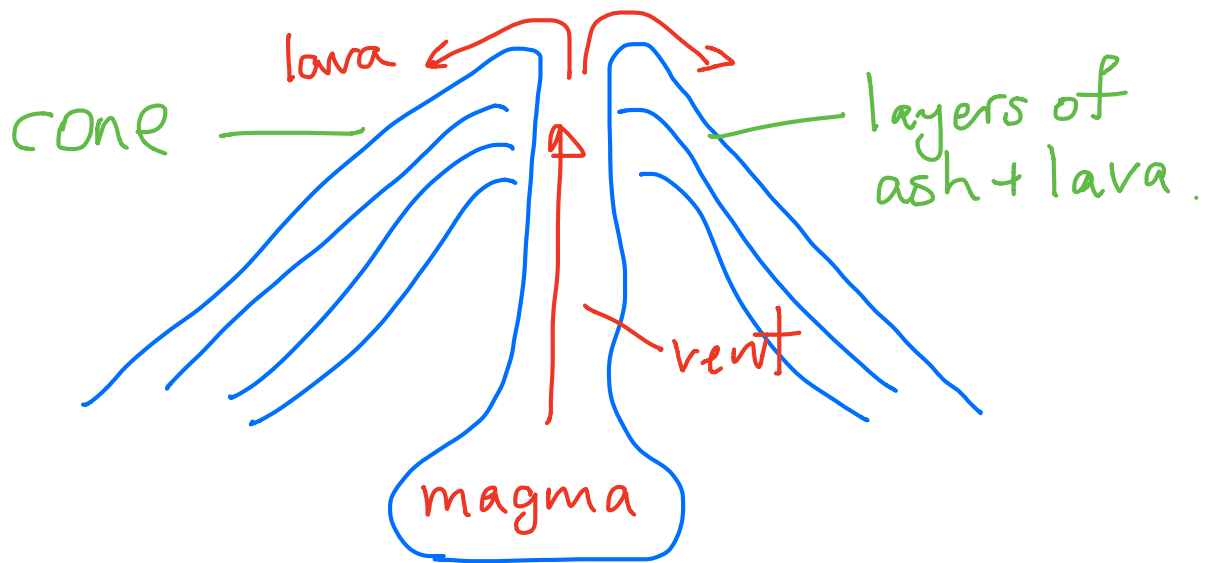
Dormant - hasn't erupted for more than 20 yrs.

Extinct - " " " thousands of years.

16. (a) How do volcanoes form?

Magma escapes to the surface through a crack.

(b) Draw a simple labelled diagram to show the major parts of a volcano.



(c) Name the three different types of volcanoes.

shield cinder cone composite cone

(d) Where on the Earth's surface are most volcanoes found? Explain why this is so.

Edge of tectonic plates.

Movement of one plate under the other forces magma upwards.

17. (a) What is a tsunami and how is it formed?

Landslip or earthquake under the ocean pushes a huge volume of water upwards.

(b) How is a tsunami different to a "normal" wave we see on our beaches?

It is a "shelf" of water with few waves.

(c) Why do tsunamis cause so much damage as they near land?

The waves move slower in shallow water. The waves behind catch up, increasing the height significantly.